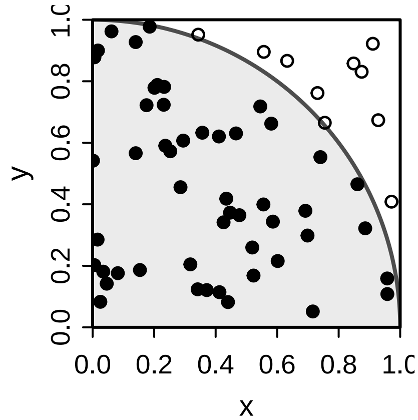
Estimating π by throwing darts

The setup

You have a square dartboard with sides of length 1. Inside it, drawn with one corner as the centre, is a quarter of a circle with radius 1. The picture shows 60 random throws.



Every throw lands somewhere in the square, completely at random: no spot is more likely than any other. Some throws land inside the quarter circle (filled dots), some land outside it (open circles).

Two formulas you already know:

Shape	Area
Square with side length of 1	$A = 1^2 = 1$
Circle with radius of 1	$A = \pi \cdot 1^2 = \pi$

What we're trying to do

Nobody has told you the value of π. All you can do is throw darts and count.

Your task. Explain how you could use a large number of random throws to *estimate* π, using only the two area formulas above and the counts of filled and open dots.

Think about:

1. The chance that a single random throw lands inside the quarter circle.
2. A formula that turns "number of throws" and "number inside" into an estimate of π.
3. A guess: if you throw 100 darts, roughly how close to π do you expect your estimate to be? What about 10,000 darts?

Then open the applet at https://driegert.github.io/demos/estimate-pi/ and see how your reasoning plays out.

Hints

Hint 1: compare areas. For a random throw, the chance of landing in a region is

$$P(\text{lands in region}) = \frac{\text{area of the region}}{\text{area of the whole square}}.$$

The square has side 1. The quarter circle has radius 1, and it is one quarter of a full circle.

Hint 2: turn the chance into a count. If you throw n darts and each has the same chance p of landing inside, then the proportion that land inside,

$$\hat{p} = \frac{\text{number inside}}{n},$$

should be close to p when n is large. Rearrange the expression for p from Hint 1 so that π is by itself on one side, then swap p for $\hat{p}$.

Hint 3: how close is close? Each throw is a yes/no trial with success probability p. You may already know a formula for the standard deviation of a sample proportion. Notice how it depends on n: to cut the typical error by a factor of 10, how many times more darts do you need?